

FORT YUKON, AK, USA

WMO: 701940

Lat: 66.567N

Lon: 145.267W

Elev: 136

StdP: 99.71

Time Zone: -9.00 (NAK)

Period: 82-19

WBAN: 26413

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -47.8	(c) -45.0	(d) -52.8	(e) 0.0	(f) -48.5	(g) -49.4	(h) 0.0	(i) -28.5	(j) 8.4	(k) -19.3	(l) 7.5	(m) -19.9	(n) 0.3	(o) 40	(p) 0.704

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	26.6	15.4	24.9	14.9	23.0	14.2	16.8	24.7	15.7	23.5	15.0	22.3	2.4	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.7	9.9	18.9	12.4	9.1	17.9	11.3	8.5	17.1	47.4	24.4	44.4	23.6	41.9	22.3	20.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.8	(b) 6.8	(c) 5.7	(e) -47.9	(f) 29.5	(g) 3.1	(h) 2.1	(i) -50.1	(j) 31.0	(k) -52.0	(l) 32.2	(m) -53.7	(n) 33.4	(o) -56.0	(p) 34.9		
(4) 7.8	(5) 6.8	(6) 5.7	(7) N/A	(8) N/A	(9) N/A	(10) N/A	(11) N/A	(12) N/A	(13) N/A	(14) N/A	(15) N/A	(16) N/A	(17) N/A	(18) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.0	-25.7	-23.5	-16.3	-4.0	8.0	15.5	16.9	12.7	5.5	-6.8	-20.4	-23.4		
	DBStd	17.22	11.66	9.14	6.91	6.68	5.04	3.34	2.89	3.60	4.36	6.81	8.51	9.04		
	HDD10.0	6008	1106	939	816	421	97	3	1	15	142	521	912	1036		
	HDD18.3	8554	1364	1172	1074	671	320	94	61	178	384	779	1162	1294		
	CDD10.0	526	0	0	0	0	36	169	215	98	8	0	0	0		
	CDD18.3	29	0	0	0	0	0	10	17	3	0	0	0	0		
	CDH23.3	333	0	0	0	0	11	112	169	33	0	0	0	8		
	CDH26.7	48	0	0	0	0	1	12	26	4	0	0	0	6		
(14) Wind	WSAvg	2.1	1.3	1.5	2.4	3.0	3.0	2.4	2.3	2.1	2.3	2.1	1.5	1.4		
(15) Precipitation	PrecAvg	395	26	21	17	12	22	40	57	60	49	35	28	29		
	PrecMax	635	69	49	68	38	55	94	110	101	108	87	78	69		
	PrecMin	281	5	1	4	2	7	11	25	7	9	10	2	8		
	PrecStd	87	17	11	13	8	11	17	20	19	25	17	19	15		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	-0.9	4.1	14.0	25.0	28.2	29.3	27.0	19.1	8.4	0.1	0.8		
		MCWB	0.7	-2.1	0.1	6.7	14.0	15.3	16.6	17.2	11.3	5.0	-5.5	0.4		
	2%	DB	2.5	-4.3	0.9	11.2	22.2	26.2	27.1	23.4	17.2	4.9	-3.9	-7.0		
		MCWB	-4.2	-5.4	-1.9	4.8	11.4	14.4	16.2	14.0	11.1	2.4	-8.3	-7.5		
	5%	DB	-7.6	-7.8	-2.5	8.6	19.1	24.3	25.3	21.2	15.0	2.7	-7.0	-10.0		
		MCWB	-9.5	-8.5	-5.1	3.3	9.7	14.1	15.9	13.5	9.5	1.1	-8.3	-10.5		
	10%	DB	-12.3	-11.4	-5.6	6.3	17.3	22.7	23.6	19.0	12.7	1.2	-8.9	-12.3		
		MCWB	-12.8	-12.1	-7.4	1.6	8.9	13.9	14.8	12.6	8.0	-0.2	-10.5	-12.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.4	-1.0	0.1	6.6	14.8	17.4	18.6	17.2	12.6	4.1	-1.7	-2.1		
		MCDB	11.5	0.2	3.9	13.4	25.8	25.6	25.5	24.1	17.3	5.8	6.3	-1.1		
	2%	WB	0.4	-4.5	-2.4	4.7	11.2	16.1	17.3	15.3	10.6	1.9	-7.8	-6.5		
		MCDB	10.1	-3.4	0.8	10.3	19.4	24.4	24.9	21.6	14.6	3.2	-6.6	-5.8		
	5%	WB	-1.6	-8.5	-5.5	2.7	9.8	15.4	16.3	13.8	9.2	0.4	-10.3	-9.7		
		MCDB	6.7	-7.9	-3.6	7.4	16.9	23.6	24.3	19.5	12.7	1.5	-9.3	-9.3		
	10%	WB	-8.1	-12.6	-7.9	0.7	8.6	14.7	15.3	12.9	7.9	-0.5	-12.4	-12.3		
		MCDB	-6.0	-12.1	-6.4	4.6	15.7	22.4	22.6	17.8	11.4	0.4	-11.4	-11.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	9.7	12.7	12.1	11.8	10.9	10.0	9.7	9.5	6.7	7.6	8.4		
		MCDBR	9.9	10.6	14.6	13.2	14.5	13.2	13.4	12.7	12.9	7.3	8.3	13.7		
	5% WB	MCWBR	6.7	10.1	13.1	8.7	7.7	5.4	6.1	7.1	9.0	5.6	9.2	9.8		
		MCWBR	7.2	11.0	15.4	13.2	13.4	12.6	12.8	11.1	11.0	6.2	9.9	9.8		
(40) Clear-Sky Solar Irradiance	taub	taub	0.165	0.231	0.252	0.328	0.339	0.362	0.381	0.368	0.316	0.250	0.163	N/A		
		taud	1.925	2.192	2.271	2.148	2.424	2.385	2.355	2.396	2.510	2.402	1.955	N/A		
	Ebn at Noon	Ebn at Noon	258	649	828	822	862	844	811	783	755	633	246	0		
		Edh at Noon	15	47	75	112	96	102	103	89	63	39	13	0		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.09	0.72	2.51	4.69	5.75	6.21	5.29	3.84	2.32	0.80	0.16	0.00		
	RadStd	RadStd	0.01	0.05	0.12	0.23	0.40	0.45	0.34	0.32	0.25	0.12	0.02	0.00		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page